

mfpml: Multi-fidelity probabilistic machine learning toolkit

Jiaxiang Yi¹ and Ji Cheng²

¹ Faculty of Mechanical Engineering, Delft University of Technology, the Netherlands ² City University of Hong Kong, Hong Kong ¶ Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Editor: [Vangelis Kourlitis](#)

Reviewers:

- [@salrm8](#)
- [@jungtaekkim](#)

Submitted: 24 December 2024

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

The mfpml (multi-fidelity probabilistic machine learning) package provides a Python platform for implementing classic single- and multi-fidelity Bayesian machine learning surrogates and applying them to Bayesian optimization. Although numerous methods have been developed in the domain of multi-fidelity machine learning ([Giselle Fernández-Godino, 2023](#)) on regression purpose, no open-source software offers a comprehensive suite of tools for multi-fidelity Bayesian optimization. This package addresses this gap by providing a platform to replicate existing single- and multi-fidelity Bayesian methods based on Gaussian process regression. Build upon on this, we provide protocols for multi-fidelity Bayesian optimization that can actively optimize a black box problem with automatically adding new design of experiments from different fidelities considering its contribution, cost, and noise. It serves as a handy tool for developing new methods in the field of multi-fidelity probabilistic machine learning and Bayesian optimization.



Figure 1: Logo of mfpml ([mfpml](#)).

Statement of Need

mfpml is written in Python and depends on a few third-party packages, such as NumPy ([Harris et al., 2020](#)) and SciPy ([Virtanen et al., 2020](#)). It includes detailed notebooks and autogenerated Sphinx documentation, allowing users to replicate existing methods and develop new ones with ease. Specifically, it provides essential modules for building machine learning models, including design of experiments, benchmark problems, models, and optimization.

Key features of mfpml include:

1. **Basic Methods:** Fundamental implementations of popular methods, such as Gaussian process regression ([Rasmussen & Williams, 2005](#)), Co-Kriging ([Forrester et al., 2007](#)), and corresponding extensions ([Han & Görtz, 2012](#)).
2. **Advanced Methods:** Advanced techniques for Bayesian optimization ([Jones et al., 1998](#)), including single-fidelity and multi-fidelity optimization.
3. **Future Development:** Ongoing work includes adding constrained optimization and multi-objective optimization methods, which will be included in future versions.

Recently, probabilistic machine learning and the corresponding Bayesian optimization frameworks have received increasing attention, where various multi-fidelity probabilistic machine learning models (Forrester et al., 2007; Han & Görtz, 2012; Rasmussen & Williams, 2005) have been employed for different learning functions, such as MF-GP-UCB (Cheng et al., 2022), BOCA (Kandasamy et al., 2016), and multi-information-source optimization (Poloczek et al., 2016), among others (Li & Li, 2024; Peherstorfer et al., 2018). These studies have established a solid methodological foundation; however, they are typically presented as standalone algorithms rather than as part of a unified and extensible software framework. These methods are particularly important for real-world simulation- or experiment-driven problems, where it is common to have multiple data sources with different fidelities, i.e., a limited number of accurate high-fidelity data points and a large amount of biased low-fidelity data. Representative applications include aerodynamic and structural design optimization (Yi et al., 2025), material modeling and discovery (Yi et al., 2026), and structural reliability analysis Yi et al. (2021), among others.

In a similar scope, several Python packages provide Gaussian process or Bayesian optimization functionality, such as GPyTorch (Gardner et al., 2021), BoTorch (Balandat et al., 2020), Ax (Olson et al., 2025), and Emukit (Paley et al., 2023), primarily based on PyTorch. Moreover, Trieste (Picheny et al., 2023) provides multi-fidelity modeling workflows within a TensorFlow-based Bayesian optimization framework. However, these frameworks primarily target single- or multi-fidelity modeling or deep Gaussian processes and are often coupled with large software dependencies. The SMT toolbox (Saves et al., 2024) offers various surrogate models with limited capability to integrate multi-fidelity data, yet it lacks support for multi-fidelity Bayesian optimization. By contrast, mfpml is designed as a lightweight and standalone toolkit that focuses explicitly on multi-fidelity Gaussian process regression and the corresponding Bayesian optimization framework. It serves both as a platform for developing and benchmarking novel multi-fidelity Gaussian process methods and as an easily deployable tool for real-world applications such as aerodynamic shape optimization, materials design, and other simulation-driven engineering problems.

Acknowledgements

We thank the Python code of conduct provided by BessaGroup and acknowledge the support from our institutions.

References

- Balandat, M., Karrer, B., Jiang, D. R., Daulton, S., Letham, B., Wilson, A. G., & Bakshy, E. (2020). BoTorch: A Framework for Efficient Monte-Carlo Bayesian Optimization. *Advances in Neural Information Processing Systems* 33. <http://arxiv.org/abs/1910.06403>
- Cheng, J., Lin, Q., & Yi, J. (2022). An enhanced variable-fidelity optimization approach for constrained optimization problems and its parallelization. *Structural and Multidisciplinary Optimization*, 65(7), 188.
- Forrester, A. I. J., Sobester, A., & Keane, A. J. (2007). Multi-fidelity optimization via surrogate modelling. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 463(2088), 3251–3269. <https://doi.org/10.1098/rspa.2007.1900>
- Gardner, J. R., Pleiss, G., Bindel, D., Weinberger, K. Q., & Wilson, A. G. (2021). GPyTorch: Blackbox matrix-matrix gaussian process inference with GPU acceleration. <https://arxiv.org/abs/1809.11165>
- Giselle Fernández-Godino, M. (2023). Review of multi-fidelity models. *Advances in Computational Science and Engineering*, 1(4), 351–400. <https://doi.org/10.3934/acse.2023015>

- 79 Han, Z.-H., & Görtz, S. (2012). Hierarchical kriging model for variable-fidelity surrogate
80 modeling. *AIAA Journal*, 50(9), 1885–1896. <https://doi.org/10.2514/1.J051354>
- 81 Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D.,
82 Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk,
83 M. H. van, Brett, M., Haldane, A., Río, J. F. del, Wiebe, M., Peterson, P., ... Oliphant,
84 T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>
85
- 86 Jones, D. R., Schonlau, M., & Welch, W. J. (1998). Efficient global optimization of expensive
87 black-box functions. *Journal of Global Optimization*, 13(4), 455–492. <https://doi.org/10.1023/A:1008306431147>
88
- 89 Kandasamy, K., Dasarathy, G., Oliva, J. B., Schneider, J., & Póczos, B. (2016). Multi-fidelity
90 bayesian optimisation with continuous approximations. *ICML*.
- 91 Li, K., & Li, F. (2024). Multi-fidelity methods for optimization: A survey. *ACM Computing*
92 *Surveys*.
- 93 Olson, M., Santorella, E., Tiao, L. C., Cakmak, S., Garrard, M., Daulton, S., Lin, Z. J., Ament,
94 S., Beckerman, B., Onofrey, E., Igusti, P., Lara, C., Letham, B., Cardoso, C., Shen, S. S.,
95 Lin, A. C., Grange, M., Kashtelyan, E., Eriksson, D., ... Bakshy, Eytan. (2025). Ax: A
96 Platform for Adaptive Experimentation. *AutoML 2025 ABCD Track*.
- 97 Paleyes, A., Mahsereci, M., & Lawrence, N. D. (2023). Emukit: A Python toolkit for decision
98 making under uncertainty. *Proceedings of the Python in Science Conference*.
- 99 Peherstorfer, B., Willcox, K., & Gunzburger, M. (2018). Survey of multifidelity methods in
100 uncertainty propagation, inference, and optimization. *Siam Review*, 60(3), 550–591.
- 101 Picheny, V., Berkeley, J., Moss, H. B., Stojic, H., Granta, U., Ober, S. W., Artemev, A., Ghani,
102 K., Goodall, A., Paleyes, A., Vakili, S., Pascual-Diaz, S., Markou, S., Qing, J., Loka, N.
103 R. B. S., Couckuyt, I., & Morter, C. (2023). Trieste: Efficiently exploring the depths of
104 black-box functions with TensorFlow. *arXiv*. <https://doi.org/10.48550/ARXIV.2302.08436>
- 105 Poloczek, M., Wang, J., & Frazier, P. I. (2016). Multi-information source optimization.
106 <https://arxiv.org/abs/1603.00389>
- 107 Rasmussen, C. E., & Williams, C. K. I. (2005). *Gaussian Processes for Machine Learning*. The
108 MIT Press. <https://doi.org/10.7551/mitpress/3206.001.0001>
- 109 Saves, P., Lafage, R., Bartoli, N., Diouane, Y., Bussemaker, J., Lefebvre, T., Hwang, J. T.,
110 Morlier, J., & Martins, J. R. R. A. (2024). SMT 2.0: A surrogate modeling toolbox with
111 a focus on hierarchical and mixed variables gaussian processes. *Advances in Engineering*
112 *Software*, 188, 103571. <https://doi.org/https://doi.org/10.1016/j.advengsoft.2023.103571>
- 113 Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D.,
114 Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson,
115 J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... SciPy
116 1.0 Contributors. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in
117 Python. *Nature Methods*, 17, 261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- 118 Yi, J., Cheng, J., & Bessa, M. A. (2025). Practical multi-fidelity machine learning: Fusion of
119 deterministic and bayesian models. <https://arxiv.org/abs/2407.15110>
- 120 Yi, J., Cheng, Y., & Liu, J. (2022). SBSC+SRU: An error-guided adaptive kriging method for
121 expensive system reliability analysis. *Structural and Multidisciplinary Optimization*, 65(5),
122 134.
- 123 Yi, J., Ferreira, B. P., & Bessa, M. A. (2026). Single-to-multi-fidelity history-dependent
124 learning with uncertainty quantification and disentanglement: Application to data-driven
125 constitutive modeling. *Computer Methods in Applied Mechanics and Engineering*, 448,

- 126 118479. <https://doi.org/https://doi.org/10.1016/j.cma.2025.118479>
- 127 Yi, J., Wu, F., Zhou, Q., Cheng, Y., Ling, H., & Liu, J. (2021). An active-learning method
128 based on multi-fidelity kriging model for structural reliability analysis. *Structural and*
129 *Multidisciplinary Optimization*, 63(1), 173–195.

DRAFT